

Project Executable Files

Date	07 July 2026
Team ID	-
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	yes
2	README / Setup Guide (instructions to run the project)	yes
3	requirements.txt / package.json / dependency file	yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	No
6	Deployed application URL (if hosted)	No(Local Flask Deployment)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
▼ SMART LENDER
  ▼ dataset
    📄 loan_prediction.csv
    📄 processed_dataset.csv
  ▼ model
    ≡ best_model.pkl
    ≡ label_encoder.pkl
    ≡ scaler.pkl
  ▼ src
    > __pycache__
    📄 data_preprocessing.py
    📄 prediction.py
    📄 train_models.py
    📄 utils.py
    📄 visualization.py
  ▼ static
    # style.css
  ▼ templates
    <> home.html
    <> predict.html
    <> submit.html
  📄 app.py
  ⓘ README.md
  ≡ requirements.txt
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted (Runs on Localhost)
Login Credentials (Demo)	Not Required
Platform / Hosting Provider	Local Flask Server
Repository Link	https://github.com/vannemreddypushpanjali/Smart-Lender-AI-ML-Project
Demo Video Link	https://drive.google.com/file/d/1Sb1QtXzV22SOO3gJ-bUKj93glzW-eTYL/view?usp=drive_link

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

1. Install **Python 3.10 or above** and **Visual Studio Code**.
2. Open **Visual Studio Code**.
3. Click **File** → **Open Folder** and open the **Smart_Lender Project** folder.
4. Open the integrated terminal by selecting **Terminal** → **New Terminal** or press **Ctrl + `**.
5. Install all the required libraries using:

```
pip install -r requirements.txt
```
6. If the model has not been trained yet, run:

```
python train_models.py
```

 This generates the trained model file (best_model.pkl).
7. Run the Flask application using:

```
python app.py
```
8. After the application starts successfully, the terminal displays:
 Running on <http://127.0.0.1:5000/>

9. Open a web browser and navigate to:

<http://127.0.0.1:5000/>

10. Click **Predict**, enter the applicant details, and click **Submit**.

- The application displays whether the loan is **Approved** or **Rejected** based on the trained machine learning model.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Uses historical dataset only	Future versions can include real-time banking data.
2	Application runs on localhost only	Can be deployed on Render, AWS, or Azure.
3	Credit score API is not integrated	Future enhancement is to integrate live credit score verification.